



▲ FAIRNESS IMPROVED

● NO ACCURACY LOSS

Executive Summary

Fairness Experiment Report - Upload 42 summarizes the current fairness state for upload 42. Accuracy changed to +0.964 and fairness score changed to +0.916. Strong fairness improvement with no measurable accuracy loss. The best balanced model is ThresholdOptimizer_mitigated_threshold with combined score 0.935.

ACCURACY

0.964

▲ +0.964

FAIRNESS SCORE

0.916

▲ +0.916

COMBINED SCORE

0.935

ThresholdOptimizer_mitigated_threshold

DISPARATE IMPACT

0.447

Ratio (DIR)

Dataset & Model Information

DATASET	MODEL	SECTIONS INCLUDED
adult.csv	dt_baseline.pkl	comparison

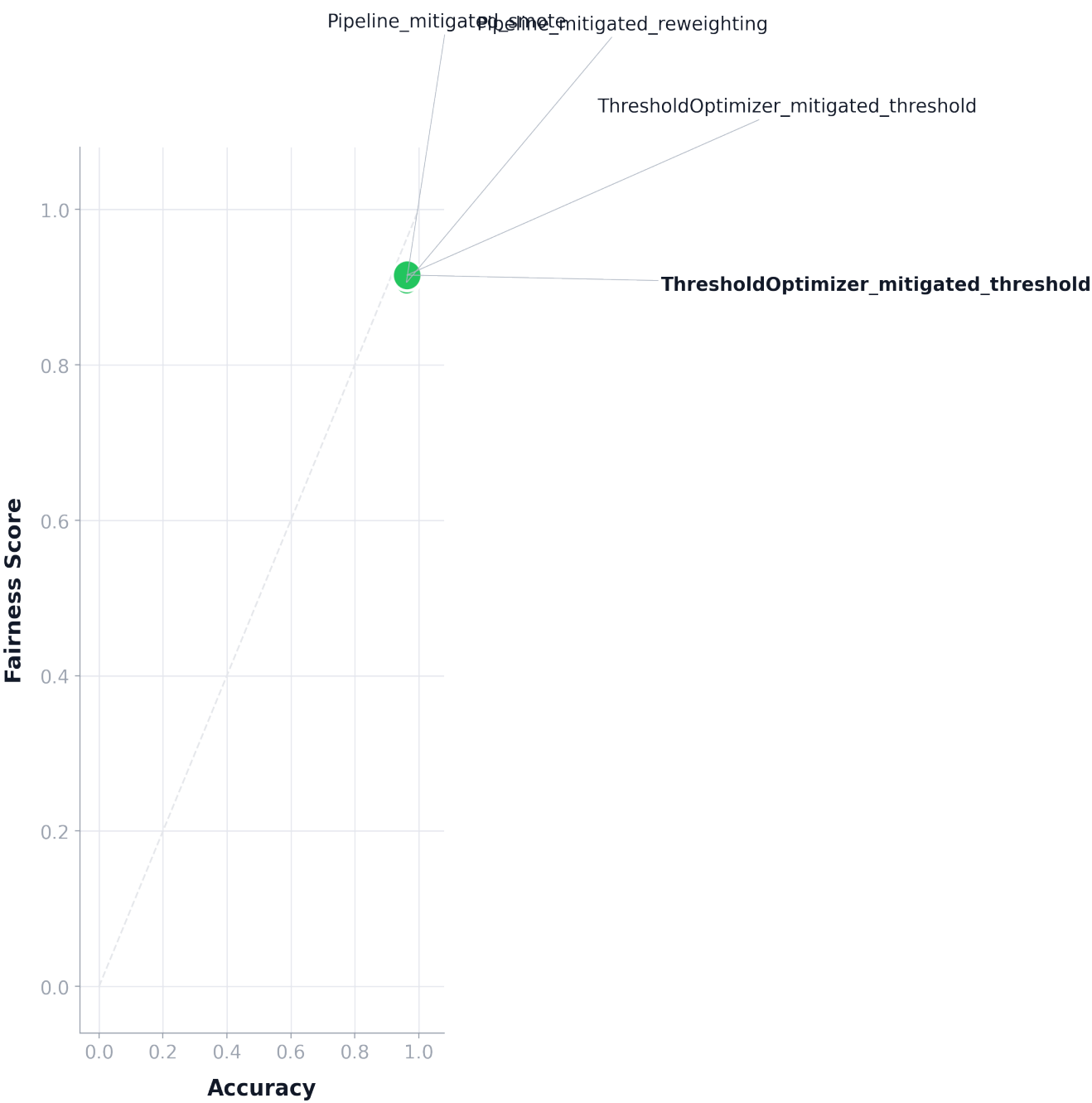
Fairness Metrics

METRIC	VALUE	INTERPRETATION
Demographic Parity Difference (DPD)	0.151	Difference in positive-prediction rates across groups; closer to 0 is more equitable.
Equalized Odds Difference (EOD)	0.018	Gap in true/false positive rates between groups; closer to 0 indicates equal error rates.
Disparate Impact Ratio (DIR)	0.447	Ratio of favorable outcome rates between groups; values near 1.0 indicate parity.

Model Comparison

Model	Accuracy	Fairness	DPD	EOD	DIR	Combined
★ ThresholdOptimizer_mitigated_threshold	0.964	0.916	0.151	0.018	0.447	0.935
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Pipeline_mitigated_reweighting	0.963	0.907	0.149	0.038	0.448	0.929
Pipeline_mitigated_smote	0.962	0.906	0.150	0.037	0.455	0.928

Accuracy vs. Fairness Trade-off



Recommendations

Fairness improved after mitigation. Consider validating and deploying the mitigated model.